Summary of Changes — Manuscript # Access-2026-23347

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Instructions for Reviewers

Due to the complexity of the IEEE Access template (custom fonts, multi-column layout, `\PARstart`), the `latexdiff` tool could not produce a compilable highlighted PDF. This document provides a structured summary of every substantive change between the original submission and the revised manuscript. All changes address reviewer comments or incorporate new experimental data.

Page and line numbers refer to the revised manuscript PDF (`llm_annotation_paper_access_revised.pdf`).

Major Changes

1. **Abstract (p. 1).** Fully rewritten with a clearer problem–method–findings–implication structure. Approximately 30% shorter than the original. The new Abstract also summarises the expanded ten-model cross-model validation and the quality-threshold experiment.

2. **Introduction, new paragraphs (p. 1–2).**
   - Formal definition of *visual anchoring*: "the degree to which a category label can be recovered from a single cropped frame without recourse to discourse context, temporal information, or speaker intent."
   - Explicit motivation for agreement filtering: explains the intuitive rationale (dual-model agreement as a reliability heuristic) before discussing its failure modes.

3. **Methods, new Pipeline Overview subsection (p. 2).** Three-stage summary of the experimental pipeline (Annotation → Filtering → Fine-tuning), with a compact enumerated list covering all design choices.

4. **Methods, Table 1–2 merged (p. 3).** The original separate tables (Dataset Overview and Per-Class Counts) are merged into a single table with per-class bounding-box counts as sub-rows.

5. **Methods, cross-model section expanded (p. 5).** The cross-model robustness check now describes ten MLLMs (expanded from five), spanning three model families (Qwen, LLaVA, Gemma) and parameter scales from 7B to 35B.

6. **Results, expanded cross-model tables (p. 7–8).** `tab:crossmodel_overall` and `tab:crossmodel_inter` now include all ten models with verified accuracy numbers. All values are sourced from actual JSONL/CSV result files or execution logs.

7. **Results, new quality-threshold experiment (p. 7).** New subsection reporting a targeted experiment with Qwen3.5-27B pseudo-labels on TeacherBehavior. Key finding: the critical annotation accuracy threshold at which pseudo-label fine-tuning surpasses the zero-shot baseline lies between 41% and 50%. Includes new `tab:quality_threshold` and `fig:quality_threshold`.

8. **Discussion, Finding 3 rewritten (p. 9–10).** Updated to incorporate the quality-threshold evidence and the finding that the selective-routing gain scales with annotator quality.

9. **Discussion, Limitations updated (p. 10–11).** The ten-model cross-model validation is now described as providing annotator-side external validity. The single-benchmark constraint is explicitly acknowledged.

10. **Conclusions rewritten (p. 11–12).** Now includes: (a) the quality-threshold finding, (b) the capability boundary on guide/answer categories, and (c) a practical annotator selection guideline.

Minor Changes (Reviewer 2, Comment 2.7)

- Long paragraphs in Introduction and Discussion split into shorter units.
- Technical terms defined at first use.
- Hedged compound sentences replaced by direct statements.
- Tables reordered and captions shortened to avoid repetition.

New Experimental Data Added

- Cross-model validation expanded from 5 to 10 models (Qwen2-VL-7B, LLaVA-1.5-7B, Qwen2.5-VL-7B, Gemma-3-27B, Qwen3.5-27B, Qwen3.5-35B-A3B, Qwen3.6-27B, Qwen3.6-35B-A3B, Gemma4-26B-A4B, Gemma4-31B).
- Qwen3.5-27B quality-threshold fine-tuning experiment on TeacherBehavior.
- All new data verified against result files on the experiment server.

Data Provenance

All numerical values in the revised manuscript can be traced to specific JSON/CSV/JSONL files in the accompanying GitHub repository (`https://github.com/zhanglizhuo/llm-annotation`). Qwen3.6-27B values are reconstructed from execution logs (result directory was lost before backup).